

Final Presentation (MIN E 630)

Underground Mining and Bulk Materials Handling

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Agenda

- ① Term paper: motivation, framing, and key takeaways
- ② Beyond the paper: learning from writing and from this course
- ③ Insights and gains from this course



About Jinpeng Zhu

- Graduate student (Civil and Environmental Engineering)
- Interest: sensing + underground mining+ decision-making
- Focus today: lifecycle-oriented sensing for cementitious ground support



Background and motivation

Term paper title: Sensing technologies for cementitious ground support systems in underground mining: advances, challenges, and future perspectives



Figure 1: The location of my term paper



Cementitious support is everywhere

Where it appears



Figure 2: Shotcrete (drifts, intersections)



Figure 3: Cast-in-place linings (shafts, chambers)



Figure 4: Cemented backfill (stopes)



Figure 5: Grouted bolts/cables

Why it is hard to manage

- Performance changes with time
- Coupled hazards environmental (stress, water, seismicity, blasting)
- Limited visibility and limited verification

Image sources:

Figure 2 source: <https://can.sika.com/en/construction/shotcrete-tunnelingmining.html>

Figure 3 source: <https://everestconstructionforms.com/access-mining-shafts/>

Figure 4 source: <https://can.sika.com/en/construction/shotcrete-tunnelingmining/mine-backfillingmbfproducts.html>

Figure 5 source:

<https://upload.wikimedia.org/wikipedia/commons/b/b9/Usgsrockbolt.jpg>



Key message and contribution

The excavation process of underground mining includes many complex procedures, harsh environments, and low visibility, so sensing technologies are necessary for improving productivity and safety.

This paper:

- Frames cementitious ground support sensing across five lifecycle phases
- Links physical variables and sensing functions to operational decisions
- Discuss why system integration (data transfer, fusion, analysis, and decision-making) is the bottleneck



Lifecycle Framing: Lifecycle-oriented questions

① Phase I - hours to days

- When can we re-enter safely?
- Did we place it well (thickness/coverage)?

② Phase II - days to weeks

- Is it carrying strength as expected?
- Are we seeing early cracking or interface issues?

③ Phase III - months to years

- Is it degrading (creep, corrosion, cracking)?
- When should we maintain, reinforce, or reassess capacity?

④ *Phase IV - Dynamic events (blasting/seismic)*

- Can the dynamic events or hazards be detected?
- Can we know what has changed due to these events?

⑤ *Phase V - Post-event assessment (inspection window)*

- What changed (geometry, cracking, hidden defects)?
- Repair, reinforce, or abandon?



Relationship between sensing and support principles

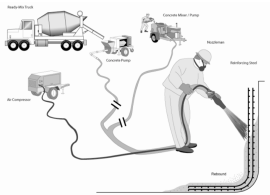


Figure 6: Cementitious systems: shotcrete wet-mix process

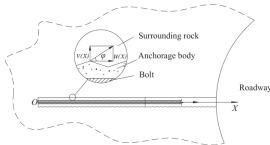


Figure 8: Grouted bolts mechanics: rock-anchorage interface schematic

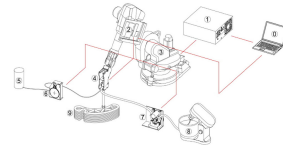


Figure 7: Cementitious systems: 3D printing schematic

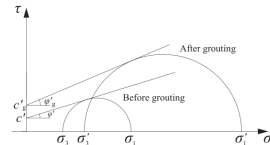


Figure 9: Grouted bolts mechanics: failure envelope before and after grouting



Lifecycle challenges (map)

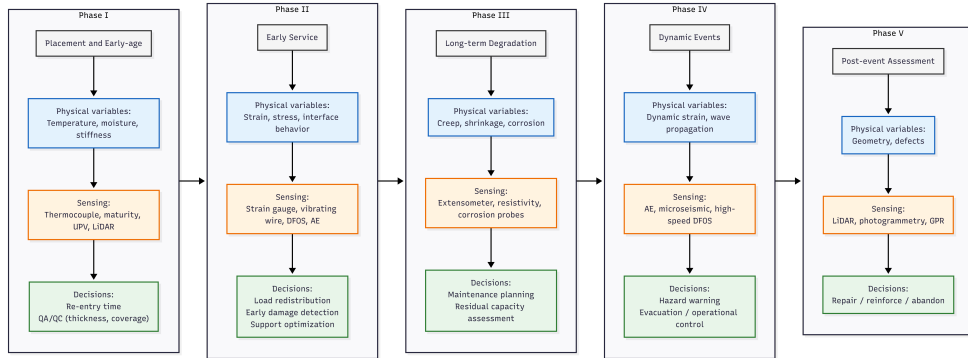


Figure 10: Lifecycle map (variables -> sensing -> decisions)



Technology selection (From sensor to decision)

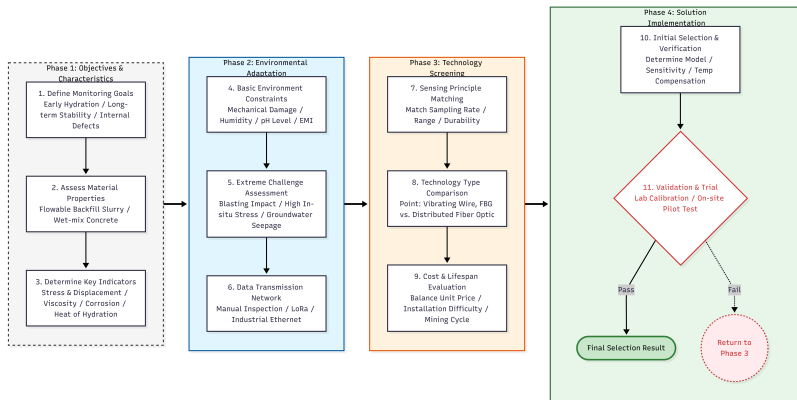


Figure 11: Technology selection workflow (goals -> constraints -> screening -> pilot)

- Thermal / maturity (temperature, early strength proxy)
- Mechanical response (stress/strain, **vibrating wire**)
- Deformation (displacement, **MDT**)
- Damage / wave-based (AE, UPV, microseismic)
- Durability / chemical (corrosion, resistivity)
- Geometry (LiDAR, photogrammetry)



Two short video: **Invisible dinosaur**, and **UPV test for concrete**, **NDT**

System Integration: communication, fusion, analysis, and action

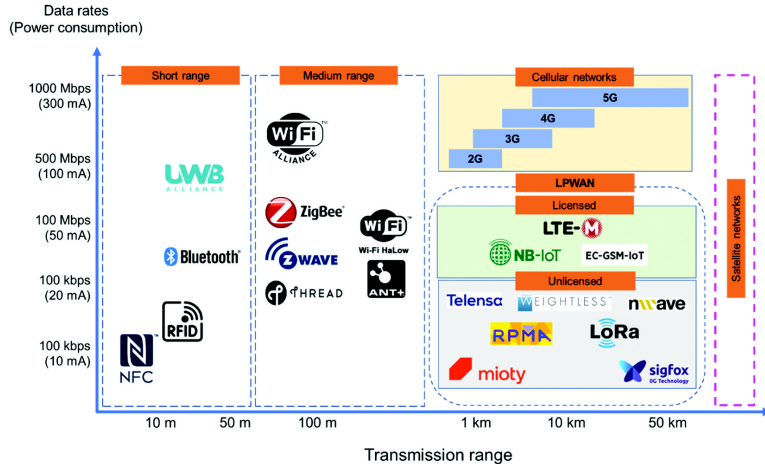


Figure 12: Comparison of wireless communication systems



Why integration is the bottleneck

- Underground comms are constrained (geometry, rock mass, no GPS)
- Data is heterogeneous (multi-modal: discrete, ascoustic, visual, etc)
- Data fusion analysis, and decision-making are based on comprehensive understanding of the system, not just anysigle sensor, or single measurement.



Challenges and future research directions

① Sensor longevity and robustness

- Sensors must survive harsh environment, such as water, blasting vibration, installation damage
- Long-term reliability in shotcrete, backfill, and grouted bolts is still a bottleneck

② Transition from monitoring to active control

- Most sites collect data but still decide by habit or experience
- We need more smarter models and auditable rules that connect and hybrid indicators to actions
- The industry don't have motivation to invest the sensing system; they only do what the regulations require

③ Towards self-assessment and self-decision

- Move from passive dashboards to structured decision support
- Use cutting-edge technologies to assist decision-making



Process Reflections & Learning from this course

Through this course, I realized that:

- ① Writing is not a easy thing, inputs and ideas are the pre-requisite
- ② Resources and support from the instructor and university were critical to our learning and growth
- ③ The really important knowledge might not come from the slides, it is from the discussions and observations with others
- ④ Industry knowledge is not always right; it's better to understand the theory and principles behind the practice, and then make your own judgment
- ⑤ Dare to challenge the status quo, which is not reasonable
- ⑥ Don't just upgrade your engine, focus on the target, and notice the environment



Closing

Thank you. Questions?

